

GRAYS REEF BUOY 41008, USA

WMO: 997010

Lat: 31.400N

Lon: 80.868W

Elev: 0

StdP: 14.70

Time Zone: -5.00 (W05)

Period: 02-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	37.2	40.7	21.4	16.1	41.3	25.5	19.5	45.8	33.0	45.8	31.5	47.0	21.6	300	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.9	86.0	80.7	84.9	79.6	84.1	78.9	81.6	83.7	80.6	83.0	80.0	82.6	10.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.9	160.6	83.3	80.1	156.0	82.6	79.2	151.5	82.1	45.2	83.9	44.2	83.1	43.4	82.4	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
30.9	26.4	24.1	DB	32.6	87.4	4.7	1.9	29.3	88.8	26.5	89.9	23.9	91.0	20.5	92.4
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	68.6	53.3	54.3	59.0	66.4	73.4	79.0	80.6	81.2	79.1	73.0	64.5	58.2
	DBStd	11.02	6.75	5.35	5.02	3.85	3.15	2.05	1.86	2.18	2.67	4.17	5.77	6.43
	HDD50	88	47	21	4	0	0	0	0	0	0	0	1	15
	HDD65	1180	364	301	191	28	0	0	0	0	0	5	74	217
	CDD50	6871	148	140	284	493	726	870	950	966	873	714	437	270
	CDD65	2487	0	0	6	70	261	420	485	501	423	255	59	7
	CDH74	19990	0	0	0	7	776	3731	5072	5473	3899	992	40	0
	CDH80	3240	0	0	0	0	5	344	973	1327	582	9	0	0
Wind	WSAvg	12.8	13.7	12.8	12.5	12.2	12.5	12.4	11.9	11.6	13.6	13.1	14.4	13.4
Precipitation	PrecAvg	52.7	4.0	3.6	4.0	3.0	3.3	6.7	6.3	6.6	5.0	4.0	2.6	3.5
	PrecMax	66.7	11.5	10.4	10.0	6.6	9.1	10.0	11.2	12.6	13.1	11.9	5.6	8.3
	PrecMin	39.9	1.1	0.9	0.6	0.8	0.7	2.3	2.6	3.7	1.5	0.4	0.4	1.2
	PrecStd	7.5	2.2	2.0	2.2	1.6	2.0	2.1	1.9	2.4	2.7	1.5	1.5	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.3	65.3	71.0	75.2	80.5	86.2	87.0	87.8	86.1	81.2	76.7	69.5
		MCWB	64.7	63.9	67.9	71.6	74.4	79.8	83.6	81.9	78.8	75.2	73.3	67.5
	2%	DB	64.8	63.7	69.1	73.5	79.2	83.7	85.3	86.3	84.3	80.2	75.3	68.5
		MCWB	63.5	62.4	66.6	71.2	73.9	78.8	80.0	80.8	78.2	74.9	73.5	66.8
	5%	DB	63.6	62.4	66.8	72.5	78.2	82.6	84.2	85.3	83.5	79.4	73.7	67.6
		MCWB	62.3	61.0	64.1	70.1	73.6	77.9	79.2	80.0	77.9	74.5	71.0	66.0
	10%	DB	61.9	61.0	65.1	71.3	77.4	81.9	83.4	84.4	82.7	78.4	72.0	66.2
		MCWB	60.3	59.4	62.3	68.6	73.4	77.4	78.6	79.2	77.2	73.5	68.8	64.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.8	65.0	69.3	73.8	77.4	80.5	83.2	83.1	81.0	78.7	75.3	68.7
		MCDB	66.0	65.1	70.3	74.0	78.0	82.2	84.7	85.4	83.1	79.6	75.5	68.8
	2%	WB	64.3	63.4	67.8	72.5	76.2	79.7	81.2	82.0	80.0	77.1	73.9	67.6
		MCDB	64.7	63.7	69.0	73.2	77.5	81.6	82.9	84.5	82.5	78.9	74.8	67.9
	5%	WB	63.0	62.0	65.4	71.1	75.3	79.0	80.4	81.1	79.2	76.0	72.6	66.2
		MCDB	63.7	62.3	66.5	72.1	77.0	81.1	82.5	83.8	82.0	78.2	74.1	66.8
	10%	WB	61.3	60.3	63.6	69.6	74.6	78.2	79.8	80.3	78.4	74.9	69.4	64.7
		MCDB	62.2	60.9	64.7	70.7	76.6	80.6	82.3	83.1	81.6	77.5	70.9	65.6
Mean Daily Temperature Range	5% DB	MDBR	7.1	6.7	6.0	5.2	4.2	3.6	3.6	3.9	4.0	4.9	6.0	6.6
		MCDBR	6.2	5.6	5.4	5.2	4.8	4.4	4.5	5.0	4.7	4.6	4.9	6.0
		MCWBR	6.6	5.8	5.1	4.7	4.5	4.2	4.7	4.5	4.0	4.6	4.8	6.5
	5% WB	MCDBR	6.1	5.2	4.9	4.0	3.7	3.5	3.8	4.4	3.9	4.3	3.7	4.9
MCWBR		6.6	5.5	4.9	4.0	3.8	4.6	4.5	4.7	3.8	5.5	4.4	5.6	
Clear-Sky Solar Irradiance	taub	0.338	0.348	0.374	0.399	0.444	0.483	0.507	0.493	0.427	0.399	0.359	0.348	
	taud	2.532	2.509	2.452	2.372	2.288	2.235	2.174	2.217	2.394	2.442	2.505	2.527	
	Ebn at Noon	282	290	288	283	270	258	251	253	268	268	272	272	
	Edh at Noon	27	30	35	39	43	45	47	45	36	32	28	26	
All-Sky Solar Radiation	RadAvg	954	1165	1547	1943	2107	2074	2056	1854	1542	1346	1054	851	
	RadStd	63	128	146	124	116	161	109	119	140	134	66	63	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (31 neighbors)	+0.95	+1.44	N/A	N/A	N/A	N/A	N/A	-135	+319	+230

Nomenclature: See separate page